

RetireWise AI

Institutional Product Requirements Document (PRD) • Version 1.0

Project: RetireWise AI	Version: 1.0 (MVP Complete) + V2 Roadmap	Classification: Open-Source / Institutional
Target Market: Pakistan WealthTech / Pensions	License: Apache 2.0 (\$0 Cost)	Status: Production Verified

1. Executive Summary & Vision

RetireWise AI is an open-source, institutional-grade retirement suitability and cash flow intelligence platform engineered specifically for the Pakistani macroeconomic environment. Designed for financial advisers, Wealth Managers, Asset Management Companies (AMCs), and Pension Fund Managers, RetireWise AI bridges the severe retirement readiness gap in Pakistan by replacing guesswork with transparent, deterministic financial mathematics and verifiable AI explanations.

THE CARDINAL SYSTEM RULE:

"The LLM explains the numbers; it does not create the numbers."

All financial mathematics, solvency years, cash flows, and suitability scores are calculated deterministically in pure Python. The AI layer acts strictly as an audit and narrative translation layer, eliminating hallucination risks entirely.

2. Problem Statement & Market Opportunity

- Retirement planning in Pakistan faces three critical structural failures that standard Western financial software fails to solve:
- **Conflation of Risk Appetite with Loss Capacity:** Advisers routinely allocate retiree portfolios to volatile equities because the client has an 'aggressive attitude,' ignoring that an imminent market crash would permanently deplete their essential living capital.
 - **Hyper-Inflation & Longevity Compounding:** With persistent double-digit inflation cycles in Pakistan (8%–15%+), static projections severely underestimate cost-of-living increases, leading to premature capital depletion in the 60s and 70s.
 - **Local Instrument Realities:** In Pakistan, retirement wealth is concentrated in illiquid property, gold, Provident Funds, Gratuities, and Voluntary Pension Schemes (VPS), requiring local tax and liquidity modeling.

3. Target User Personas

User Persona	Key Needs & Objectives	Platform Value Delivered
Financial Adviser / Wealth Planner	Institutional suitability compliance, Fact-Find automation, audit-proof recommendations, and PDF report delivery.	Reduces assessment turnaround from 3 hours to 3 minutes with automated conflict detection and client-ready PDFs.
Pre-Retiree / Client (Age 45–65)	Clarity on whether savings will outlive them, realistic inflation expectations, and emergency reserve safety.	Clear, plain-language executive summaries explaining safe withdrawal rates and stress test vulnerabilities.
Institutional AMC / Pension Fund	Regulatory compliance (SECP), anti-mis-selling safeguards, standardized risk-capacity scoring, and white-labeling.	Standardized open-source calculation framework with full traceability and offline deployment capability.

4. Core Functional Modules & Specifications

4.1. Attitudinal Risk vs. Objective Capacity for Loss

The system evaluates clients along two independent orthogonal axes to produce a 2D Suitability Matrix:

- 1. **Attitudinal Risk Tolerance (0–100):** 8-question psychometric instrument scoring market volatility comfort, time horizon perception, and panic-selling triggers.
- 2. **Objective Capacity for Loss (0–100):** Deterministic algorithmic formula evaluating liquid emergency runway, portfolio income dependency, and debt encumbrance ratio.
- 3. **Suitability Conflict Detector:** Automated trigger identifying dangerous divergences (e.g. Persona C: High Risk Appetite 90 vs Low Capacity 20), outputting compulsory warnings and adviser action checklists.

4.2. Longevity Cash Flow & 5 Stress Scenarios

Simulates year-by-year cash flows across the entire planning horizon (up to age 85+). In each cycle, investment returns grow capital while inflation-indexed living expenses and guaranteed incomes determine net withdrawals. Every client profile is subjected to 5 distinct macroeconomic stressors:

Scenario Name	Macroeconomic / Life Shock Applied	Key Vulnerability Exposed
1. Base Case	Baseline expected inflation and return rates	Baseline capital depletion runway
2. High Inflation Shock	+5.0% persistent inflation above baseline	Purchasing power erosion and rapid reserve drain
3. Early Market Drawdown	-4.0% annualized returns across years 1–5	Sequence-of-returns risk during initial retirement
4. Medical Expense Shock	PKR 2,500,000 lump sum healthcare cost at age 70	Liquid buffer adequacy and catastrophic erosion
5. Early Retirement Shock	Forced retirement 3 years ahead of schedule	Loss of peak accumulation years + lengthened horizon

4.3. Provider-Agnostic AI Narrative Layer

The AI layer consumes condensed JSON payloads (<1,500 tokens) to generate plain-language executive summaries, audit flags, and client-facing discussion guides. It dynamically supports Groq (Llama 3.3 70B, Qwen 27B), Google Gemini, and OpenAI, alongside a built-in deterministic offline engine that functions with zero external dependencies.

5. Technical Architecture, Security & Privacy

Layer	Technology Selection	Rationale & Security Controls
Computation Engine	Python 3.10+, Pydantic V2	Strict deterministic typing, zero math drift, 100% test coverage.
User Interface	Streamlit, Plotly Interactive Charts	Fast responsive B2B dashboard, dynamic fact-find editor.
Document Generator	ReportLab Community Edition	Direct binary PDF creation, client-side download, audit-compliant.
Database	SQLite (Migratable to PostgreSQL)	Zero-configuration persistence, isolated client case storage.
Secrets & Privacy	Environment Isolation + DOM Masking	Keys never rendered in HTML DOM; gitignored local credentials.

6. Release Roadmap & Milestones

Milestone	Target Features & Capabilities	Status
Version 1.0 (MVP)	Deterministic calculation engines, Pakistan Fact-Find, 5 stress scenarios, ReportLab 12-section PDF, Streamlit UI, 15/15 tests passing.	COMPLETED & LIVE
Version 2.0 (Planned)	Vectorized Monte Carlo (1,000 runs), Dynamic Digital Twin What-If sliders, Pakistan Shariah Allocator (Sukuks, VPS, KMI-30), Client Portal, FastAPI backend.	IN DESIGN / SPRINT READY
Version 3.0 (Future)	Multi-tenant AMC & Bank portals, SECP compliance surveillance dashboard, role-based access control (RBAC), multi-adviser branch hierarchies.	Roadmap Backlog
Version 4.0 (Enterprise)	REST API Suite: Headless Risk Profiling API, Capacity Engine API, Core Banking Integration Microservices, CRM connectors.	Roadmap Backlog

7. Key Performance Indicators (KPIs)

- **Mathematical Accuracy & Determinism:** 100% deterministic calculation reproducibility across all client portfolios.
- **Assessment Turnaround Speed:** Complete Fact-Find to multi-scenario stress test & PDF generation in under 60 seconds.
- **Mis-Selling Protection:** 100% capture rate of Risk-Capacity mismatches before product recommendations.
- **Open-Source Accessibility:** 100% free-tier and zero-cost stack compatibility for individual advisers and boutique wealth planners.